

Continuous Virus Inactivation: How to Generate a Plug Flow

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To realize continuous integrated manufacturing of biopharmaceuticals, all steps of production have to be operated in a continuous mode. Virus inactivation is a crucial step in this process, and due to standard procedures and a fixed residence time, it is particularly challenging to realize. Two articles in this issue present possible solutions to this problem.

Continuous integrated manufacturing will radically change the way how industry will produce biopharmaceuticals in future. However, the way toward this ambitious goal is not yet paved. In a stellar paper Konstantin Konstantinov and his group at Sanofi presented a convincing analysis of the economic benefit of integrate continuous biomanufacturing (<http://www.engconf.org/17184-2/integrated-continuous-biomanufacturing-iii/>).^[1] He also received the “Integrated Continuous Biomanufacturing Award” for his outstanding engagement to develop this field at the 3rd Integrated Continuous Biomanufacturing Conference (ICB III) held in Cascais, Portugal.^[2] Although “truly continuous manufacturing” has been already demonstrated at laboratory and pilot scale, there are yet a lot of problems to be solved. One is continuous virus inactivation.

In case of manufacturing with mammalian host cells, such as Chinese Hamster Ovarian (CHO) cells, two virus clearance steps based on orthogonal principles are mandatory.^[3,4] Currently, manufacturing of antibodies is a combination of chromatography, filtration, and virus inactivation/clearance steps. The cell culture broth generated usually by fed batch culture is harvested by filtration or centrifugation or a combination of both and then captured by affinity chromatography. To render such a process into a “truly continuous manufacturing” process, when chromatography is performed

either in a cyclic pseudo-continuous operation, or in best case, in counter-current operation. Then, the protein harvest of the individual unit operations is stored in surge tanks before loaded onto the next step.^[5]

Incubation at low pH is a preferred virus inactivation step, because it is powerfully clearing retroviruses such as Murine Leukemia Virus. In batch operation, the

protein solution is usually incubated for 1 h, although the viruses are immediately cleared. The safety margin by incubation of 1 h seems to be carved in stone, although it is not based on scientific principles. Nevertheless, in order to avoid further discussion industry renders a 1-hour batch incubation into a continuous process with a residence time of 1 h. The other reason for a on hour residence time, is to enlarge the operation window and cover worst case conditions typically considered during a failure mode analysis when a process is characterized.^[6] Now the question arises how to design a reactor with a narrow residence time distribution to warrant the 1-hour incubation, but also to avoid over-incubation with the consequence of product loss. The plug flow reactor is the solution, but how can one achieve plug flow? In an empty tube a turbulent flow is required to achieve a reasonable plug flow, but this would require extremely large reactors to meet the target. The second reason why turbulent flow is not desired is to reduce the pressure drop, in particular if multiple unit operations are connected. Turbulent flow results in a large pressure drop across a unit operation, thus it is not recommended. Therefore, operation must be conducted under laminar flow conditions. For example, Orozco et al.^[7] compared different concepts for continuous virus inactivation using tubular reactors.

In this issue, two concepts for such reactors are presented.^[8,9] In one case the principle of Dean vortices is applied.^[8] In a curved channel a secondary flow is generated under a certain flow regime which leads to efficient mixing but also stabilizes the plug flow, because the typical parabolic flow pattern observed in tubes and capillaries is not generated. This is also called the coiled flow inversion reactor.^[8] Amarikwa et al. at Boehringer solved this problem by using the principles of a jig in the box reactor which is very similar to the coiled flow inversion reactor. This reactor was originally designed for heat exchange and as an inline mixer and has been also successfully applied for virus inactivation^[10] and protein precipitation.^[11] The article by Amarikwa et al. shows an interesting methodology, which allows a very accurate and sensitive determination of plug flow **Figure 1**.

Gillespie et al.^[9] from MilliporeSigma use a tubular reactor to generate a plug flow. They present a general concept, how such a

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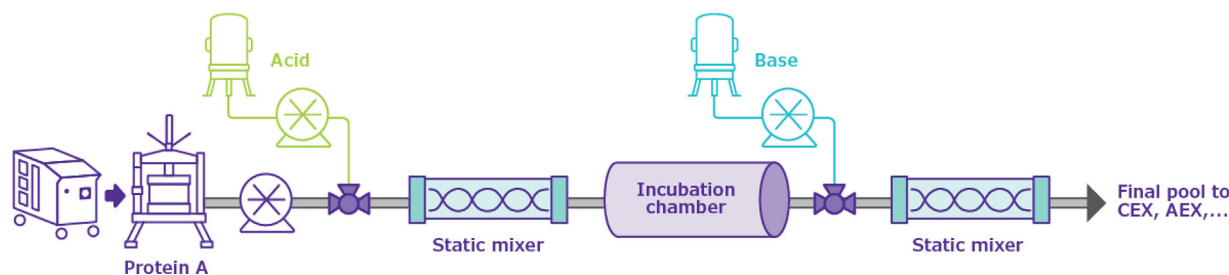


Figure 1. Concept for implementation of continuous in-line virus inactivation. Reproduced with permission. Copyright 2018, Wiley-VCH.

reactor could be implemented in a continuous manufacturing train. The tubular reactor has been also already successfully tested for continuous flocculation and precipitation with in-built static mixers.^[12–14]

In this respect, both reactors are perfect for continuous operation, because the mass flow can be kept constant. Unfortunately, Gillespie et al.^[9] do not disclose the nature of reactor. In any case, the work is useful, because it teaches how to implement such reactors. I am convinced that these works will be scaled up and operation will be demonstrated at pilot and full scale soon.

Conflict of Interest

The authors declare no commercial or financial conflict of interest.

Keywords

antibody, continuous, IgG, integrated biomanufacturing, low pH, virus

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